Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production Python engineering, specializing in **LLM evaluation infrastructure, agentic systems, and production AI deployment**. Built AgentProbe — an open-source evaluation platform with a 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval quality, completeness, compliance), CI/CD quality gates, ChromaDB RAG observability, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama). Built LeaseI — an agentic RAG compliance pipeline (93.6% precision, 87.9% F1) with a pass@k eval harness and LLM-as-judge scoring. Shipped Entrata's first production AI pipeline (PyTorch + OpenAI API) across 1.6M real financial records with output validation, quality gates, and compliance controls. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Trainee → Associate Engineer → Software Engineer

- Shipped Entrata's **first production AI pipeline** across 1.6M real financial records — output validation, quality gates, and compliance controls in place at launch; later productized into a customer-facing feature (Python, PyTorch, OpenAI API).
- Achieved **100% PHI compliance** across all client data by migrating to AWS S3 with Libsodium-encrypted pipelines — zero security incidents; built and operated data pipelines across S3, EC2, RDS, and Glue for 1M+ property records.
- Cut deployment times **40%** and sustained **99.5% uptime** across 6+ months by shipping 20+ production modules (Python, PostgreSQL, MySQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users.
- Eliminated **80% of manual operational workload** with zero compliance violations by automating LIHTC, Section 8/HCV, and HUD workflows — income eligibility verification, AMI calculations, and recertification pipelines across 100K+ unit portfolios.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · Anomaly Detection

- Achieved **$R^2 = 0.977$** regression accuracy across 190+ countries by building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets; surfaced anomalies manual review missed across 3 research pipelines by applying Isolation Forest.
- Improved research comprehension **40%** across two cohorts by designing faculty leadership dashboards in Tableau, ggplot2, and Quarto.

PROJECTS

AgentProbe: LLM Evaluation & Agentic AI Observability Platform

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · LLM-as-Judge · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Blocked failing agent runs before production by building a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval quality, completeness, compliance) with CI/CD quality gates — every commit evaluated automatically.
- Deployed a **15-endpoint FastAPI backend** running 100% locally at zero cost by building a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), tool calling, and inter-agent state handoffs.
- Delivered **trace-level observability** across concurrent agent test suites — per-run metrics, regression drift, retrieval quality, and latency — by building a React dashboard backed by a ChromaDB RAG pipeline.

LeaseI: Agentic RAG Pipeline with Pass@k Evaluation Harness

Python · LangGraph · LLaMA · Transformers · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs by building an agentic LangGraph RAG pipeline — deterministic rules engine keeps every decision fully auditable.
- Reached **pass@1 81.7% and pass@3 88.3%** by implementing a pass@k eval harness with multi-trial sampling, regression gate, and LLM-as-judge scoring of accuracy and reasoning quality.

AI on FHIR: NLP Pipeline for Regulated Clinical Records

Flask · spaCy · BERT · Transformers · Embeddings · LLMs · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time **~60%** and enforced HIPAA-compliant RBAC across structured medical records by building an NLP platform (spaCy, BERT, embeddings) translating natural language queries (EN/ES) to FHIR APIs — demonstrates production AI with safety and guardrail constraints.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

LLM & Eval: LLM-as-Judge, LLM Evaluation, Eval Frameworks, Pass@k Harnesses, Hallucination Detection, Retrieval Quality Metrics, CI/CD Quality Gates, Prompt Engineering, LLM Inference, Multi-Provider LLM Routing, OpenAI API, Anthropic API, Groq, Ollama

Agents & RAG: LangGraph, LangChain, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, ChromaDB, FAISS, Vector Databases, Semantic Search, Embeddings, Agent Observability, FastAPI, REST APIs

ML & NLP: PyTorch, Scikit-learn, spaCy, BERT, Transformers, XGBoost, Isolation Forest, Feature Engineering, Pandas, NumPy, Hugging Face

Languages: Python, SQL, PL/SQL, JavaScript, TypeScript, R, Bash, HTML/CSS

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, SQLite, RabbitMQ, Git, Linux, Agile/Scrum (120+ sprints)